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# A Meta-Analysis of Single-Case Research on Embedding Interests of Individuals With Autism Spectrum Disorder

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**Abstract**

Restricted interests of individuals with autism spectrum disorder (ASD) can act as a barrier to learning important life skills. Embedding the interests of individuals into their learning environment can act to establish motivation to participate. A meta-analysis of 20 single-case studies on embedding interests of individuals with ASD was conducted to determine overall effects and potential moderating variables. Data were aggregated across domains of variables including dependent variables, participant characteristics, and learning contexts. The correlation between self-stimulatory or ritualistic behaviors and primary outcomes was evaluated. Publication bias and consistency of the Tau/Tau-*U* nonoverlap effect size with visual analysis were tested via moderator analyses. The literature suggests that embedding interest for learners with ASD can either be beneficial or distractible. Effects ranged from negative to strong, with several moderators identified. Practitioners should take caution when using this intervention with students who display self-stimulatory or ritualistic behaviors when their interests are present.

**Keywords**

autism, interest, preference, embed, engagement

Autism spectrum disorder (ASD) is characterized by deficits in areas of social communication as well as the presence of repetitive or restricted behaviors and interests (RRBIs; American Psychiatric Association, 2013). RRBIs are displayed in several ways, including stereotypic motor movements or speech, insistence on sameness, sensory hypo- or hyperactivity, and restricted interests (also known as circumscribed or perseverative interests). The interests of individuals with ASD tend to be more intense and differ in content relative to the interests of typically developing individuals (Anthony et al., 2013). Restricted interests are often expressed through restricted or repetitive object use, preoccupations with certain attributes of stimuli, or fascinations for certain topics or activities (Bishop et al., 2013; Bishop, Richler, & Lord, 2006; Bruckner & Yoder, 2007). Restricted interests can become more apparent with age and perhaps more interfering or resistant to change (South, Ozonoff, & McMahon, 2005).

Restricted object use can interfere with important sources of alternative stimulation and be at odds with the development of pivotal skills (Bruckner & Yoder, 2007). Targeting the reduction of restricted interests may successfully reduce related problem behavior but may not readily treat the intensity of restricted interests (Boyd, Woodward, & Bodfish, 2011). Alternatively, the interests of individuals with ASD can be utilized in strengths-based approaches to intervention. One such strengths-based approach is embedding interests into the learning environment. Interventions that embed interests involve incorporating individually identified interests (i.e., stimuli that an individual tends to interact with at their leisure) into aspects of a curriculum or activity to promote participation in low-preferred or low-probability activities (Mancil & Pearl, 2008). Embedding interests might be useful because preferred stimuli will temporarily alter the value of neutrally preferred stimuli and evoke engagement. This is in line with the theoretical construct of motivating operations (Michael, 1993, 2000; Sundberg, 1993). Systematic evaluation of outcomes resulting from embedding interests as an independent variable can elucidate the evidence base and

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theoretical base of this intervention, which may have applied value for individuals with ASD.

Three meta-analytic reviews have been conducted on incorporating interests for individuals with ASD (Dunst, Trivette, & Hamby, 2012a, 2012b; Raab & Dunst, 2007). Dunst et al. (2012a) conducted a meta-analysis of 14 single-case studies with children with ASD using the intraindividual point biserial correlation coefficient. Dunst et al. (2012b) meta-analyzed 24 group-based and single-case studies with children with ASD using the Cohen's *d* effect size. Raab and Dunst (2007) aggregated 25 group-based and single-case studies with children both with and without disabilities using the Cohen's *d* effect size. Dunst et al. (2012a, 2012b) found that effects tended to have larger improvements with age and children with milder symptoms of ASD produced slightly larger effects relative to those with moderate or severe symptoms.

There are limitations in these reviews to consider. Included studies were not appraised for quality and they had substantial methodological differences. For instance, studies incorporating situational interests were included, defined as characteristics of the social or nonsocial environment (e.g., sights and sounds) that invite attention or engagement with material or people (e.g., social stories, music, animal-assisted therapy). Situational interests were not necessitated to be determined individually (i.e., on a case-by-case basis), and these treatments targeted different outcomes than embedding interests (i.e., reductions in challenging behavior or engagement with the "situational interest" itself versus alternative, low- or neutrally preferred stimuli). Given these concerns, a review is warranted specifically on embedding interests that incorporates techniques to evaluate internal validity.

Two additional reviews have evaluated the topic of embedding interests for individuals with ASD (Gunn & Delafield-Butt, 2016; Ninci, Rispoli, Burke, & Neely, 2018). Gunn and Delafield-Butt (2016) evaluated studies using the perseverative interests of individuals with ASD by either providing them as reinforcers or embedding them as an antecedent approach. These studies were exclusively peer-reviewed, but they were not evaluated based on quality standards, and the review was descriptive in nature. The authors reported generally favorable outcomes of 20 studies on the topic of utilizing restricted interests, although many studies included in this review did not experimentally isolate the sole independent variable of embedding interests. Ninci et al. (2018) conducted a review to investigate the quality of interventions that embed interests of individuals with ASD and found mixed outcomes overall with the most support for teaching social communication skills. This review included 20 studies—both dissertations and peer-reviewed publications. The mixed outcomes of this quality review indicated a need for an analysis of potential moderators through meta-analysis to determine the variables contributing to the range of effects, including negative effects detected through visual analysis.

Publication bias, or the tendency to only report or recommend publication of favorable results, is another threat that is increasingly recognized to be reflected in behavior analytic literature bases (Shadish, Zelinsky, Vevea, & Kratochwill, 2016). Sham and Smith (2014) conducted a review of a common intervention technique and found that published studies tended to include cases with larger effect sizes than that found in unpublished dissertations. Although dissertations do not undergo the conventions of a rigorous peer-review, they can still be subjected to an appraisal of methodological quality. An inclusive approach to reviews could, at least in part, mitigate the effects of publication bias and help in determining potential moderators in meta-analyses of single-case research (Shadish et al., 2016).

The majority of the literature on embedding individual interests is single-case research. To improve the validity of single-case meta-analysis, Burns (2012) suggested controlling for confounds of poorly defined constructs, conducting exhaustive searches to prevent the potential effects of publication bias, requiring minimum quality indicators as criteria for including studies, and addressing statistical validity of effect sizes used. No one statistical method comprises all considerations that take place in the process of visual analysis (e.g., evaluation of level, trend, immediacy of effect, consistency of effects across replications). Visual analysis is nuanced, considerate to all patterns of single-case data, and it is used to detect functional relations (Ledford & Gast, 2018). Although visual analysis is the primary method of data evaluation for single-case designs, it does not produce a standard metric to interpret the magnitude and generality of effects. The credibility of inferences from statistics can be validated by comparing effect size outcomes to visual analysis ratings.

Previous studies have demonstrated that the Tau or Tau-*U* nonoverlap effect size showed general consistency with visual analysis indicators (Ledford, King, Harbin, & Zimmerman, 2018; Ninci et al., 2015). The Tau nonoverlap effect size is a robust measure of nonoverlap of all pairs of data points between two conditions (Parker, Vannest, Davis, & Sauber, 2011b). A number of meta-analyses have used the nonparametric statistics of Tau or Tau-*U* (that controls for baseline trend) and nonoverlap of all pairs (NAP, which differs from Tau/Tau-*U* in scale; Parker, Vannest, & Davis, 2011a) to measure effects (e.g., Ledford et al., 2018; Ninci et al., 2015; Roth, Gillis, & DiGennaro Reed, 2014). These methods fit the assumptions of single-case design and consider all data (Parker et al., 2011a, 2011b). A limitation to the Tau/Tau-*U* nonoverlap metric is that it is restricted to representing nonoverlap of data points between phases with control for baseline trend in the direction of desired change, if applicable. Therefore, this metric and others should be interpreted as a supplement to the visual analysis process.

### Purpose

The purpose of this review is to extend the previous reviews by (a) refining inclusion criteria; (b) evaluating same, similar, and new potential moderators; (c) evaluating publication bias; and (d) evaluating results based on quality and visually analyzed indicators. Research questions included the following:

**Research Question 1:** What are the overall effects of embedding interests on outcomes for individuals with ASD for dependent variables targeted for increasing and for those targeted for decreasing?
**Research Question 2:** What are the effects for each included dependent variable?
**Research Question 3:** What is the correlation between self-stimulatory or ritualistic behaviors and primary outcomes?
**Research Question 4:** What are the effects of potential moderators including (a) participant characteristics, (b) learning contexts, (c) publication types, and (d) visual analysis of evidence categories for cases according to the What Works Clearinghouse (WWC, 2014)?

## Method

### Search Procedures

Three databases were systematically searched including Education Resource Information Center (ERIC), PsycINFO, and ProQuest Dissertations and Theses Full Text in the fall of 2017. Publications and dissertations were included. All pairs of terms from two sets of words were searched: (a) *autis**, "*pervasive developmental*," or *Asperger** and (b) *prefer**, *reinforc**, *interest**, *perseverat**, *circumscribed, ritual**, *obsessi**, and *restrict**. Terms from word set "b" were restricted in the search to be included within the title of retrieved articles. Terms from the first word set, or word set "a," were also paired with the terms "*preferred materials*" or "*traditional materials*," without a search restriction for these terms to be included in the title. Through these procedures, 1,574 references with duplicates removed were identified.

*Title and abstract screening.* Through reviewing titles and abstracts, articles were excluded if they (a) were not in English, (b) were a literature review, (c) were descriptive (e.g., book reviews, chapters), or (d) did not include an educational/environmental intervention with measures of student/client behaviors (e.g., assessment or survey studies, medication trials, animal studies). There were 644 studies that advanced for further review.

*Full-text screening.* Studies were completely evaluated to identify if they (a) isolated embedded interests and (b) included an individual with ASD. Embedded interest interventions fix preferred stimuli into an activity to increase interaction with alternative stimuli. Studies attempted to isolate this by comparing standard, neutral, or low-preferred arrangements to high-preferred arrangements. The intensity of participant interests could not be categorized given the detail of some studies, but all studies had to use observational information or indirect descriptive reports (e.g., interview with stakeholders) to determine participants' individualized interests. Studies were not included that only used a preference assessment with stimuli determined a priori. Dissertations were not retrieved if they were placed under an embargo for a period of time. Two group-based studies (i.e., Donaldson & Olswang, 2007; Nadig, Lee, Singh, Bosshart, & Ozonoff, 2010) were identified and excluded from further inclusion to use a common metric for this meta-analysis. Single-case studies were included that made a systematic attempt to replicate effects in a graphed experimental design at least twice (e.g., a multiple baseline design across two participants replicates A-B phases twice through a stagger in initial implementation of the intervention phase, an alternating treatment design rapidly alternates between conditions to replicate effects). Thirteen single-case studies were identified to be included.

*Ancestral search.* Iterative ancestral searches were applied in the summer of 2018, including a Google Scholar database search of references citing the included studies and a reference list review for each study. Article titles were reviewed and if necessary, the abstract and full text were reviewed based on the inclusion and exclusion criteria. The database search resulted in an additional six studies and the reference list search resulted in an additional one study, making a total of 20 included studies.

*Hand search.* Hand searches from the first to most recent issues of *Journal of Positive Behavior Interventions* (January 1999—April 2018) and *Journal of Autism and Developmental Disorders* (January 1971—June 2018) were conducted. We selected these two journals because they included the most articles retrieved that fit the study inclusion criteria. Editorials, book reviews, letters, and messages were not counted in the total articles list. Articles, original papers, and brief reports were evaluated from the journals. A total of 457 titles were screened out of 78 issues in *Journal of Positive Behavior Interventions*, which yielded four abstract reviews and no full-text reviews (four retrieved articles were already included in this review). A total of 4,421 titles were screened out of 306 issues in *Journal of Autism and Developmental Disorders*, which yielded four abstract reviews and four full-text reviews (two articles were already included in this review). The hand searches did not yield any additional works meeting the inclusion criteria for this study.

*Interrater reliability (IRR) on screening.* Another rater used the criteria to independently evaluate 626 randomly selected references from 1,574 (40%) for the title and abstract screening and 222 randomly selected references from 644 (34%) for the full-text screening. IRR was computed by dividing the rater agreements by the agreements plus disagreements and obtaining a percentage through multiplying by 100. There was 94% agreement on general inclusion or exclusion of studies for the title and abstract screening. There was 97% agreement across studies on the inclusion criteria items for the full-text screening. Disagreements at each step in the screening process were discussed until a consensus between the two raters was met.

## WWC Standards

Cases from studies were included that systematically manipulated the independent variable of embedding interests with individuals with ASD. According to WWC (2014), this is the first design standard for inclusion for single-case research. Specific participants were eliminated here if their interests were reported to deteriorate through the study (two participants from Adams, 1999) or interests were not informed by original observational or indirect reports (two participants from Agarwal, 2012). This was done to better isolate the independent variable, as an evident inconsistency in identified interests across preference assessment methods or time was considered to be a major threat to internal validity.

The remaining WWC design standards were applied to determine if studies should be further analyzed for evidence standards. Specifically, studies were analyzed to determine if they met minimum criteria for interobserver agreement (i.e., IOA collected for 20% of sessions; IOA at or above 80% or .60 for Kappa), attempts to replicate effects (at least three), and numbers of data points per phase (multiple-baseline and reversal designs necessitate five data points to be coded as "Meets Design Standards" and three to be coded as "Meets Design Standards with Reservation" alternating treatment designs necessitate five data points to be coded as "Meets Design Standards" and four to be coded as "Meets Design Standards with Reservation").

Per the WWC standards, cases considered as "Meets Design Standards" or "Meets Design Standards with Reservation" were visually analyzed within baseline and intervention phases for predictability, level, trend, and variability. A conclusion on the presence of "Basic Effects" between phases or conditions was determined per case (i.e., participant) with consideration to level, variability, trend, immediacy of effects, consistency of effects across same phases, and overlap between phases. This does not necessarily indicate experimental control, depending on the design; but it does represent indicators that are integral to visual analysis. This analysis of basic effects between phases was a step used in a visual analysis coding protocol developed by Maggin, Briesch, and Chafouleas (2013) based on procedures described in the WWC standards for single-case research. To be inclusive given the small pool of studies and determine the statistical validity of the results (i.e., congruence of visual analysis indicators and effect sizes), the visual analysis component of the WWC evidence standards was used as a tool to reliably code all cases from included studies where researchers manipulated the independent variable. It is important to note that this is an adaptation made from the original criteria exclusively for the analysis of "Basic Effects," but further analysis of "Overall Evidence" did not include designs with cases that did not meet all of the design standards (i.e., overall evidence codes did not vary from the WWC standards). Cases with as few as two replication attempts were visually analyzed to determine the presence or absence of basic effects.

Studies that met all WWC design standards were evaluated on overall evidence. These studies met design standards or met them with reservations by manipulating the independent variable, meeting minimum criteria for interobserver agreement, demonstrating at least three attempts to replicate effects, and including minimum numbers of data points per phase. Overall evidence codes represent both the design rigor and experimental control (defined as three total replications of effect with at least three effects to each noneffect) for designs. Designs identified to have strong (i.e., at least three replications of effect, no noneffects, and "Meets Design Standards") and moderate (i.e., "Meets Design Standards with Reservation" and/or some noneffects, with at least three replications of effect per noneffect) evidence were aggregated together to be compared with designs with no evidence (i.e., fewer than three replications of effect or fewer than three replications of effect per noneffect) as well as designs that did not meet all minimum design standards.

*IRR on WWC standards.* A random selection of 11 (55% of 20) single-case studies was evaluated for IRR on WWC design standards. These studies included 61 (52% of 117) dependent variables within cases. Two independent raters agreed for an average of 94% (range, 80%–100% within design standards) of 366 agreement opportunities. Regarding WWC evidence standards, the two raters practiced together with one study and the second rater then independently evaluated seven (35% of 20) studies with 39 (37% of 105) dependent variables within cases. Raters agreed for 97% of opportunities in classifying basic effects and 92% of opportunities in classifying overall evidence.

## Variable Coding

Dependent variable categories evaluated included social communication behaviors, task-engagement/accuracy/productivity, positive affect, disruptive or off-task behavior,

and self-stimulatory or ritualistic behavior. Social communication behaviors and task engagement/accuracy/productivity were considered primary measures because they indicated engagement with alternative stimulation sources. Social communication behaviors included a variety of behaviors requiring the involvement of social partners such as social play, joint attention, and scripted conversation. Task-engagement/accuracy/productivity variables included academic (e.g., social studies, reading comprehension), preacademic (e.g., matching), functional play, and task-analyzed skills. Teaff (2001) included data on disruptive and desirable behavior, but because these were the inverse of one another, only desirable behavior was included. Positive affect, disruptive or off-task behavior, and self-stimulatory or ritualistic behavior were evaluated to supplement primary measures. Looking at targets (Naoi, Tsuchiya, Yamamoto, & Nakamura, 2008) and proximity (Talebi, 2007) were each measured once and were not included.

Age categories included 2 to 5 years or preschool-aged, 5 to 10 years of elementary-aged, and 11 to 21 years or secondary/adolescent-aged. For functioning-level categories, severe autism was assigned for participants described as having ASD and (a) severe language delays, (b) intellectual disability, (c) a reported IQ below 70, or (d) a delay of approximately 2 or more years as determined by adaptive behavior or age-equivalent scores. Autism was coded for participants described as having ASD or autism, without meeting criteria for the other categories. High-functioning autism/Asperger syndrome was coded for participants described as having relatively "high-functioning" autism, Asperger syndrome, or when participants were said to be fully on their grade level or have an average to high IQ.

Learning contexts included client/student-led versus instructor-led categories. Involvement in target behaviors were voluntary, or not under the context of an instructor demand, for the client/student-led category. If only encouragement or initial prompting in a session with fading was used, the study was classified as client/student-led. When there were specific demands, prompts, or expectations held for participants throughout the baseline and intervention, the variable was categorized as instructor-led.

Finally, studies were coded as either peer-reviewed or dissertation to determine publication bias through a moderator analysis. If dissertations produced a lower effect size with statistical significance in comparison with peer-reviewed publications, an indicator of publication bias would be demonstrated.

*IRR on variable coding.* A second evaluator determined the accuracy of ratings made by a primary evaluator to obtain IRR for eight (40% of 20) randomly selected studies. The second evaluator determined if there was an agreement or disagreement with each code by referencing the article. There were 44 of 105 (42%) dependent variables within cases analyzed across five primary moderator categories. The second rater agreed with the primary rater for 100% of opportunities.

## *Data Extraction and Analysis of Effects*

Data extraction was completed by hand to estimate values on the *y* axis in correspondence with values in each phase or condition on the *x* axis (data extractions with reliability estimates and effect size calculations can be accessed by contacting the first author). Hand extractions were done using Microsoft® Excel with screenshots of graphs and a feature to create straight lines to indicate where a given data point is along *y* axis. Low-preferred conditions were considered as baseline or control and high-preferred conditions were considered as intervention. Generalization and maintenance data were excluded. Some studies had unique conditions for which analyses were adapted. Baker (2000) and Baker, Koegel, and Koegel (1998) included a variety of baseline conditions, some of which included prompting. Because the intervention conditions in these studies initially included prompting, the baselines were considered to commence when prompting sessions were finished to isolate the intervention alone. R. L. Koegel, Dyer, and Bell (1987) used one B-A-B-A-B design and the last phase was not included because it had no comparison. This study also measured social avoidance, the inverse of which was used and considered social engagement. Finally, a participant in R. L. Koegel et al. (2012) began taking medication midway through intervention, and these sessions were not included in the analyses as a new independent variable was added, and it was suggested that this may have confounded the results. Marshall (2017) evaluated different levels of the independent variable—no embedded interests, infrequently embedded interests, and frequently embedded interests; outcomes from the infrequently embedded interests condition were not included in these analyses. Baseline phase data points representing no embedded interests were aggregated with the intervention phase data points as the procedures were the same, and this produced a more equal number of data points across the compared conditions. Watkins et al. (2017) included generalization baseline probe and programming conditions to use the intervention across peer partners, so these data were aggregated with the primary peer data in each condition.

The Tau/Tau-*U* nonoverlap metric (i.e., Tau-*U*) was computed by comparing baseline to intervention or alternating treatment conditions (Parker et al., 2011b). Computations were made using the calculator available at www.singlecaseresearch.org (Vannest, Parker, & Gonen, 2011). Trend control (Tau-*U*) was applied in instances that baseline trend was detected through visual analysis (10 phase contrasts). Trend control was not applied for alternating treatment designs because trend in the control condition could likely

occur as a result of interaction effects. The Tau-*U* nonoverlap statistic is interpreted as a percentage of data showing improvement or nonoverlap under intervention conditions, with values ranging from −1 to 1 (Parker et al., 2011b). It has no interpretation benchmarks and results can be interpreted given contextual considerations (Vannest & Ninci, 2015). To be consistent with previously published meta-analyses using similar metrics (e.g., Tincani & De Mers, 2016), results were interpreted as weak for effects and confidence intervals within the ± 0.00 to 0.65 range, moderate within the ± 0.66 to 0.92 range, and strong within the ± 0.93 and above range.

Effects for dependent variables in studies were evaluated with confidence interval ranges set at 95%. Moderator analysis results were evaluated in terms of effect size magnitudes with nonoverlap in confidence intervals ranges. Confidence interval ranges were set at 83.4% upper and lower limits to interpret nonoverlap in forest plots. At these limits, nonoverlap in ranges is approximately representative of $p < .05$ (Payton, Greenstone, & Schenker, 2003).

Primary outcomes were the only variables analyzed for the moderator analyses of age, functioning level, and learning context. Effects intended for increasing (i.e., social communication behaviors, task-engagement/accuracy/productivity, positive affect) were included for the moderator analyses of publication type, basic effects, and overall evidence. Effects intended for decreasing (i.e., disruptive or off-task behavior, and self-stimulatory or ritualistic behavior) were only analyzed in a dependent variable moderator analysis and analysis of the correlation between self-stimulatory or ritualistic behavior and primary outcomes.

*IRR on data extraction.* Data extraction was analyzed for accuracy by a second rater for 11 (55%) randomly selected studies with 25 (49%) participants and 1,193 (45% of 2,660) data points. Codes for each data point were compared to graphs to obtain a percentage of agreements from the total. There were 16 disagreements across 1,193 data points resulting in a mean of 99% (range, 87%–100% within studies) agreement. Discrepancies were analyzed until the raters reached consensus.

## Results

Table 1 presents the Tau-*U* nonoverlap effect sizes and confidence intervals (95%) for dependent variables of studies. Moderator analysis codes across dependent variables within cases can be accessed by contacting the first author. Of 48 participants with effects intended for increasing among 20 studies, 11 participants from six studies also included effects intended for decreasing. Of 147 total contrasts, 116 were intended for increasing and 31 were intended for decreasing. There were 37 targeted dependent variables intended for increasing and seven intended for decreasing. Effects intended for increasing resulted in a weak effect size of 0.55 (95% CI = [0.49, 0.61]; *SE* = 0.03). Effects intended for decreasing resulted in a weak effect size of −0.07 (95% CI = [–0.16, 0.03]; *SE* = 0.05).

### Analyses of Dependent Variables

Of 116 contrasts with dependent variables intended for increasing, 70 (60%) targeted social communication behaviors, 30 (26%) targeted task-engagement/accuracy/productivity, and 16 (14%) targeted positive affect. There was nonoverlap in confidence interval ranges between the primary outcome categories of social communication behaviors, with moderate effects (Tau-*U* = 0.79; 83.4% CI = [0.74, 0.85]), and task-engagement/accuracy/productivity, with weak effects (Tau-*U* = 0.10; 83.4% CI = [0.03, 0.17]). Positive affect behaviors were weak to moderate (Tau-*U* = 0.73; 83.4% CI = [0.60, 0.85]). Of 31 total contrasts intended for decreasing, 15 (48%) targeted disruptive or off-task behavior, and 16 (52%) targeted self-stimulatory or ritualistic behavior. Disruptive or off-task behavior resulted in weak effects (Tau-*U* = −.30; 83.4% CI = [–0.40, –0.19]) and self-stimulatory or ritualistic behavior resulted in weak effects (Tau-*U* = 0.08; 83.4% CI = [0.00, 0.17]), demonstrating no overlap in confidence interval ranges. All dependent variable categories had variable effects among phase contrasts, ranging from weak to strong. For two dependent variable categories, there were multiple phase contrasts with moderate to strong negative effects (i.e., going in the direction of unintended change); these categories were task engagement/accuracy/productivity and self-stimulatory or ritualistic behavior.

Figure 1 presents a correlation analysis between self-stimulatory or ritualistic behavior and primary outcomes (i.e., social communication behaviors and task-engagement/accuracy/productivity). Four single-case studies with eight participants and 18 primary dependent variable phase contrasts measured self-stimulatory or ritualistic behavior with primary outcomes. The analysis produced a moderate negative Pearson correlation coefficient ($r = -.73$). Increases in self-stimulatory or ritualistic behavior were associated with decreases in primary outcomes.

### Analyses of Participant Characteristics

Figure 2 presents the moderator analysis for age categories (i.e., preschool, elementary, or secondary/adolescent). Ages ranged from 2 to 18 years among 48 participants. Nine (20%) participants were classified as preschool-aged, 18 (40%) as elementary-aged, and 18 (40%) as secondary/adolescent-aged. Ages of three participants (with six phase contrasts) were unclear and not included in this analysis. Of 94 contrasts for primary dependent variables, 20 (21%) were classified as preschool-aged, 31 (33%) as elementary-aged,

**Table 1.** Tau/Tau-U Effect Sizes and CI Ranges for Dependent Variables Within Studies.

| Article | Target behavior, Number of Contrasts | Tau-*U* | 95% CI | |
|---|---|---|---|---|
| | | | LL | UL |
| Adams (1999) | *Task score* (TEAP), 9 | −0.03 | −0.21 | 0.15 |
| | *Off-task* (DOT), 9 | 0.04 | −0.15 | 0.22 |
| | *Self-stimulatory behavior* (SSRB), 9 | −0.01 | −0.20 | 0.17 |
| Agarwal (2012) | *Scripted responses* (SOCCOM), 1 | 0.16 | −0.13 | 0.45 |
| | *Unscripted responses* (SOCCOM), 1 | 0.21 | −0.08 | 0.50 |
| | *Card reader acquisition* (TEAP), 1 | −0.13 | −0.43 | 0.17 |
| Baker (2000) | *Social play* (SOCCOM), 3 | 0.99 | 0.71 | 1.00 |
| | *Joint attention* (SOCCOM), 3 | 0.97 | 0.69 | 1.00 |
| | *Affect* (AFFECT), 3 | 0.81 | 0.52 | 1.00 |
| | *Ritualistic behavior* (SSRB), 3 | −0.43 | −0.71 | −0.15 |
| Baker et al. (1998) | *Social play* (SOCCOM), 3 | 1.00 | 0.52 | 1.00 |
| | *Affect* (AFFECT), 3 | 0.90 | 0.41 | 1.00 |
| Boyd, Alter, & Conroy (2005) | *Social initiations* (SOCCOM), 1 | 1.00 | 0.15 | 1.00 |
| Boyd, Conroy, Mancil, Nakao, & Alter (2007) | *Social interactions* (SOCCOM), 3 | 0.79 | 0.44 | 1.00 |
| Clarke et al. (1995) | *Disruptive behavior* (DOT), 2 | −0.98 | −1.00 | −0.58 |
| | *Desirable behavior* (SOCCOM), 2 | −0.07 | −0.47 | 0.33 |
| | *Productivity* (TEAP), 2 | 0.89 | 0.49 | 1.00 |
| El Zein, Solis, Lang, & Kim (2016) | *Reading comprehension* (TEAP), 1 | 0.64 | 0.15 | 1.00 |
| | *Retell* (TEAP), 1 | 0.64 | 0.15 | 1.00 |
| R. L. Koegel et al. (1987) | *Social avoidance*[a] (SOCCOM), 6 | 1.00 | 0.56 | 1.00 |
| R. L. Koegel et al. (2012) | *Engaged with peers* (SOCCOM), 5 | 1.00 | 0.70 | 1.00 |
| | *Initiations with peers* (SOCCOM), 5 | 1.00 | 0.70 | 1.00 |
| R. L. Koegel, Kim, Koegel, & Schwartzman (2013) | *Engaged with peers* (SOCCOM), 7 | 0.91 | 0.66 | 1.00 |
| | *Initiations with peers* (SOCCOM), 7 | 0.88 | 0.63 | 1.00 |
| L. K. Koegel, Vernon, Koegel, Koegel, & Paullin (2012) | *Engaged with peers* (SOCCOM), 4 | 0.96 | 0.65 | 1.00 |
| | *Social initiations* (SOCCOM), 4 | 0.96 | 0.65 | 1.00 |
| Marshall (2017) | *Reading fluency* (TEAP), 2 | 0.22 | −0.20 | 0.64 |
| | *Oral retell* (TEAP), 2 | 0.06 | −0.36 | 0.48 |
| | *Reading comprehension* (TEAP), 2 | 0.49 | 0.07 | 0.91 |
| Morrison & Rosales-Ruiz (1997) | *Stereotypic behavior* (SSRB), 2 | 0.75 | 0.52 | 0.97 |
| | *Correct responses* (TEAP), 2 | −0.82 | −1.00 | −0.60 |
| Naoi et al. (2008) | *Joint attention* (SOCCOM), 3 | 0.52 | 0.06 | 0.98 |
| | *Affect* (AFFECT), 3 | 0.53 | 0.06 | 0.99 |
| Neely, Rispoli, Camargo, Davis, & Boles (2013) | *Challenging behavior* (DOT), 4 | −1.00 | −1.00 | −0.65 |
| | *Academic engagement* (TEAP), 4 | 1.00 | 0.65 | 1.00 |
| Talebi (2007) | *Social engagement* (SOCCOM), 3 | 1.00 | 0.61 | 1.00 |
| | *Social initiations* (SOCCOM), 3 | 0.88 | 0.48 | 1.00 |
| | *Affect* (AFFECT), 3 | 0.72 | 0.33 | 1.00 |
| Teaff (2001) | *Desirable behavior* (TEAP), 2 | 0.86 | 0.33 | 1.00 |
| Vismara & Lyons (2007) | *Joint attention* (SOCCOM), 4 | 0.77 | 0.37 | 1.00 |
| | *Affect* (AFFECT), 4 | 0.61 | 0.22 | 1.00 |
| Watkins et al. (2017) | *Stereotypy* (SSRB), 2 | −1.00 | −1.00 | −0.47 |
| | *Functional play* (TEAP), 2 | 1.00 | 0.47 | 1.00 |
| | *Social engagement* (SOCCOM), 2 | 1.00 | 0.47 | 1.00 |

*Note.* CI = confidence interval; LL = lower limit; UL = upper limit; SSRB = self-stimulatory or ritualistic behaviors.
[a]Effect size represents the inverse of measures.

and 43 (46%) as secondary/adolescent-aged. Effects by age categories ranged from weak to moderate with progressive increases by age category and no overlapping in confidence interval ranges from preschool to secondary/adolescent-aged and elementary to secondary/adolescent-aged. Preschool-aged resulted in weak effects (Tau-*U* = 0.34;

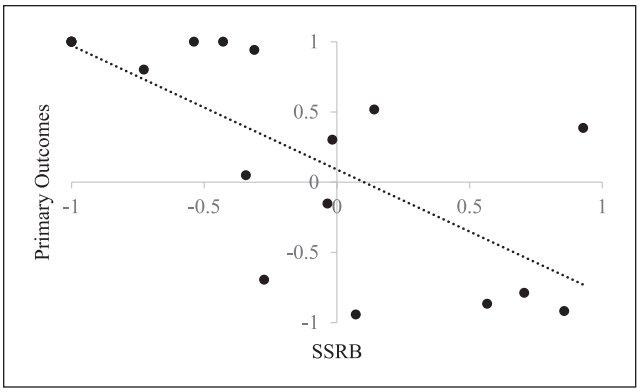



**Figure 1.** Moderate negative correlation ($r = -.73$) between Tau/Tau-*U* nonoverlap scores of SSRB and primary outcomes.
*Note.* SSRB = self-stimulatory or ritualistic behaviors.

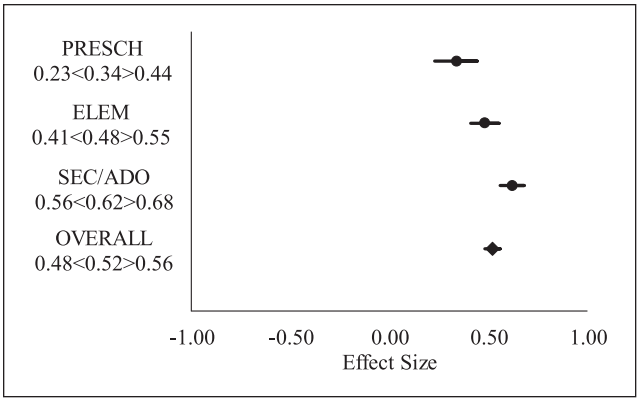



**Figure 2.** Age levels targeted in single-case studies with confidence intervals set to 83.4% upper and lower limits.
*Note.* ELEM = elementary; PRESCH = preschool; SEC/ADO = secondary/adolescent.

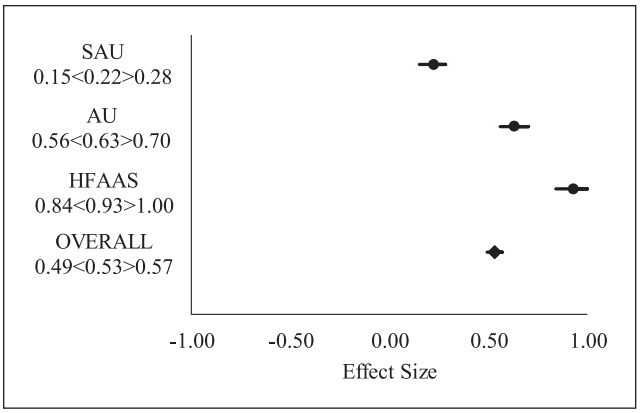



**Figure 3.** Functioning levels targeted in single-case studies with confidence intervals set to 83.4% upper and lower limits.
*Note.* AU = autism; HFAAS = high-functioning autism/Asperger syndrome; SAU = severe autism.

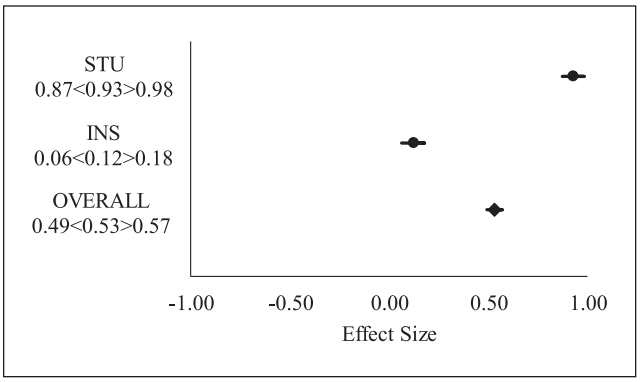



**Figure 4.** Contexts targeted in single-case studies with confidence intervals set to 83.4% upper and lower limits.
*Note.* INS = instructor-led; STU = student-led.

83.4% CI = [0.23, 0.44]), elementary-aged in weak effects (Tau-*U* = 0.48; 83.4% CI = [0.41, 0.55]), and secondary/adolescent-aged in weak to moderate effects (Tau-*U* = 0.62; 83.4% CI = [0.56, 0.68]).

Figure 3 presents the moderator analysis for functioning level (i.e., severe autism, autism, or high-functioning autism/Asperger syndrome). Sixteen (33%) participants were classified with severe autism, 17 (35%) with autism, and 15 (31%) with high-functioning autism/Asperger syndrome. Of 100 contrasts for primary dependent variables, there were 38 (38%) for severe autism, 34 (34%) for autism, and 28 (28%) for high-functioning autism/Asperger syndrome. Effects ranged from neutral to high with no overlaps in confidence interval ranges from severe autism to autism and autism to high-functioning autism/Asperger syndrome. Severe autism resulted in weak effects (Tau-*U* = 0.22; 83.4% CI = [0.15, 0.28]), autism in weak to moderate effects (Tau-*U* = 0.63; 83.4% CI = [0.56, 0.70]), and high-functioning autism/Asperger syndrome in moderate to strong effects (Tau-*U* = 0.93; 83.4% CI = [0.84, 1.00]).

## *Analysis of Learning Context*

Figure 4 presents the moderator analysis for learning context (i.e., client/student vs. instructor-led). Thirty-three (69%) participants received client/student-led interventions and 15 (31%) received instructor led-interventions. Of 100 contrasts for primary dependent variables, there were 62 (62%) for client/student-led and 38 (38%) for instructor-led. Client/student-led resulted in moderate to strong effects (Tau-*U* = 0.93; 83.4% CI = [0.87, 0.98]) with no overlap in confidence interval ranges in comparison with instructor-led, with weak effects (Tau-*U* = 0.12; 83.4% CI = [0.06, 0.18]).

### Analysis of Publication Type

Publication types (i.e., peer-reviewed versus dissertation) were evaluated as a moderator to test for publication bias. Of 116 unique contrasts for effects intended for increasing, there were 87 (75%) from peer-reviewed articles and 29 (25%) from dissertations. There were five (25%) dissertations and 15 (75%) peer-reviewed articles. Peer-reviewed resulted in weak to moderate effects (Tau-$U$ = 0.70; 83.4% CI = [0.65, 0.75]) with no overlap in confidence interval ranges in comparison with dissertations, with weak effects (Tau-$U$ = 0.25; 83.4% CI = [0.18, 0.32]). Dissertations were identified to have primarily no evidence (i.e., met design standards but did not demonstrate experimental control) according to the WWC standards. Sixteen (57%) dependent variables targeted within cases of dissertations demonstrated no evidence, whereas eight (29%) demonstrated strong or moderate evidence and four (14%) did not meet design standards. There were 53 (69%) dependent variables targeted within cases of peer-reviewed articles that demonstrated strong or moderate evidence, 21 (27%) that did not meet design standards, and three (4%) that met design standards demonstrated no evidence.

### Analyses of WWC Standards

Two WWC standards (i.e., basic effects, overall evidence) were evaluated to test the statistical validity of outcomes. Of 116 unique total contrasts for effects intended for increasing, there were 93 (80%) with a presence of basic effects and 23 (20%) with an absence of basic effects. The presence of basic effects variable resulted in moderate to high effects (Tau-$U$ = 0.90; 83.4% CI = [0.85, 0.95]) with no overlap in confidence interval ranges in comparison with the absence of basic effects variable, with weak negative effects (Tau-$U$ = −0.15; 83.4% CI = [–0.22, –0.08]).

There were 18 of 20 (90%) studies that included designs coded as meets design standards or meets Design Standards with reservations, according to the WWC design standards (WWC, 2014). Fifteen of the 18 studies evaluated for evidence standards included designs with strong or moderate evidence. Of 116 unique total contrasts for effects intended for increasing, there were 68 (59%) with strong and moderate evidence, 14 (12%) with no evidence, and 34 (29%) that did not meet design standards. Effects with strong or moderate evidence resulted in moderate to high effects (Tau-$U$ = 0.91; 83.4% CI = [0.86, 0.97]) with no overlap in confidence interval ranges in comparison with effects with no evidence, with weak effects (Tau-$U$ = −0.09; 83.4% CI = [–0.17, 0.00]). Studies not meeting minimum quality design standards resulted in weak effects (Tau-$U$ = 0.35; 83.4% CI = [0.27, 0.42]) that fell between the other two variables with no overlap in confidence interval ranges. There were 31 participants who demonstrated strong or moderate evidence of embedding interests for at least one dependent variable targeted for increasing. Eight participants demonstrated no evidence for at least one dependent variable targeted for increasing.

## Discussion

The purpose of this meta-analysis was to determine the overall effects and moderators of embedding interests for individuals with ASD. Primary outcomes included social communication behaviors and task-engagement/accuracy/productivity. Although both positive and weak or negative effects were found for each of these variables within studies, social communication behaviors more consistently demonstrated strong, positive results. Side-effect variables included positive affect, disruptive or off-task behavior, and self-stimulatory or ritualistic behavior. Individuals with ASD generally tended to have more positive affect behavior in the embedded interest conditions. Both disruptive or off-task behavior and self-stimulatory or ritualistic behavior included a wide range of effects within studies and the averages were affected by outliers. Four studies in the meta-analysis measured self-stimulatory or ritualistic behavior as an ancillary variable to primary outcomes and a correlation analysis resulted in an inverse relation. That is, single-case studies measuring and noting increases in repetitive behaviors when interests were embedded were associated with no effects or worsened outcomes on primary measures (Adams, 1999; Morrison & Rosales-Ruiz, 1997). On the contrary, Baker (2000) and Watkins et al. (2017) found self-stimulatory or ritualistic behavior to decrease and primary outcomes to increase across participants.

Literature shows that embedding interests can either evoke or abate engagement with alternative stimulation sources. It could be that when restricted interests of individuals with ASD are embedded into the learning environment, there is an increased likelihood that the stimuli could be distracting. This is supported by demonstrations of increased stereotypical behaviors in correspondence to negative primary outcomes (Adams, 1999; Morrison & Rosales-Ruiz, 1997) and decreased stereotypical behaviors in correspondence to positive primary outcomes (Baker, 2000; Watkins et al., 2017). Nadig et al. (2010), a group-based study on this topic, also found negative effects of embedding interests when restricted and repetitive interest severity was high. Individuals with more stereotypical behavior appear to have more difficulty with devoting attention to alternative stimuli during embedded interest interventions. RRBIs such as stereotypic or repetitive motor movements, vocal responses, and self-injurious behavior are often self-stimulatory or self-serving responses that are reinforced automatically (Ringdahl, Wacker, Berg, & Harding, 2001). Children with ASD who have few socially mediated reinforcers and more automatically reinforcing

behaviors tend to make less gain through early intervention (Klintwall & Eikeseth, 2012). This is likely because socially mediated reinforcement can be easily used to evoke desirable behaviors and reduce maladaptive behaviors, whereas automatically reinforced behaviors may compete with other sources of stimulation (Klintwall & Eikeseth, 2012).

Although the dependent and independent variables in previous meta-analyses were broader than in the current review, outcomes among commonly analyzed moderators tended to concur. Similar to previous reviews in this area, effects tended to show more improvement with age (Dunst et al., 2012a, 2012b). The current review also produced similar findings to these previous reviews in that participants with milder symptoms of ASD tended to make more treatment gains than participants with moderate to severe symptoms. Although restricted interests may be present more commonly among individuals on the spectrum that have higher nonverbal IQs, repetitive use of objects is more common in individuals with lower nonverbal IQs (Bishop et al., 2006). Stereotypic behaviors often include repetitive motor movements with specific or preferred stimuli, and these behaviors are generally associated with less treatment gains (Klintwall & Eikeseth, 2012). Therefore, this meta-analysis is consistent with previous research that might lead to a hypothesis that embedded interest interventions can cause increased stereotypical behaviors in individuals with moderate to severe ASD.

Another moderator analyzed was the learning context. Given the widely varied results for this moderator, it appears that the nature of the task may influence embedded interest outcomes. Embedded interest interventions with instructor demands in place resulted in weak overall effects. Embedded interest interventions that took place with no consistent instructor demands in place resulted in moderate to strong effects. Challenging behaviors can sometimes occur when ritualistic activities with interests are interrupted (e.g., Rispoli, Camargo, Machalicek, Lang, & Sigafoos, 2014); this could influence behavioral patterns during embedded interest interventions that are led by the instructor. Alternatively, naturalistically situated contingencies promote shared control to increase motivation (R. L. Koegel & Mentis, 1985). Although participant-led activities can promote some indicators of engagement, those outcomes may not necessarily equate to prosocial or appropriate behavior (Nadig et al., 2010).

Publication type was evaluated as a moderator to test publication bias. The results suggest a record of publication bias in the literature by showing positive effects to have a higher likelihood of publication in a peer-refereed journal. Analyses from dissertations commonly met design standards, but demonstrated no evidence, whereas peer-reviewed articles most frequently met design standards and demonstrated strong or moderate evidence. In addition, more analyses from peer-reviewed articles were rated to not meet design standards in comparison with dissertations; this may be because several studies predated literature that established standards for study and design rigor. This justifies the inclusion of dissertations in this review. Publication bias is a common issue across fields of study. The current study replicates findings from another retrospective analysis method used to test publication bias in a behavior analytic single-case research-base (Sham & Smith, 2014).

Two of the WWC standards were tested as moderators to assess the statistical validity and internal validity of outcomes. Methods and findings are similar to previous meta-analyses using the Tau and Tau-*U* nonoverlap effect sizes in comparison with WWC standards (Ledford et al., 2018; Ninci et al., 2015). Specifically, these effect sizes showed general consistency with dichotomous ratings on the presence versus absence of basic effects between phases or conditions as well as the presence versus absence of experimental control in designs that met design standards. Studies that did not meet design standards produced stronger effects than those that did meet design standards, but did not demonstrate evidence.

### *Limitations*

There are limitations present in this meta-analysis. Outcomes are skewed with no representation of preschoolers with high-functioning autism/Asperger syndrome. However, the validity of the functioning-level categories is questionable because some studies lacked detailed participant information. The outcomes are also skewed in that activities led by students versus instructors were more common among studies with participants categorized with high-functioning autism/Asperger syndrome. Although social communication behaviors resulted in the strongest effect as a primary dependent variable, it was more commonly targeted among this subpopulation of individuals associated with high effects. Therefore, it is difficult to attribute moderating effects to any one of these variables analyzed. In addition, there was high heterogeneity in the overall effects intended for both increasing and decreasing. Perhaps, removing outliers or looking within specific dependent variable categories to test moderators would result in different findings. One potential concern related to IRR is that study variable codes and data extractions were not assessed by two independent raters; rather, a second rater referenced the article and codes of the first rater to determine agreements and disagreements. Another potential concern is that data entry into the web-based calculator and the respective calculator estimates were not assessed for IRR. Finally, because there was evidence of publication bias and 29% of the contrasts did not include sufficient indicators of internal validity, results should be interpreted with caution.

### Future Research

Future research should be conducted to evaluate the hypotheses presented in this meta-analysis with precision. It is unclear if results vary due to the primary dependent variables, participant characteristics, and/or learning contexts. Similar to findings reported by Ninci et al. (2018), there is considerable support for studies targeting social communication behaviors. Nine studies in this review measured indicators of task-engagement/accuracy/productivity, and this support is limited due to variations in study quality and effects. Therefore, future research should continue to evaluate task-engagement/accuracy/productivity as a primary dependent variable alongside side-effect measures.

A number of studies in this meta-analysis show support for primary outcomes to generalize to the original activity (e.g., Baker, 2000; Baker et al., 1998; Naoi et al., 2008; Vismara & Lyons, 2007). Future studies should review these outcomes and isolate generalization experimentally. It is possible that embedding interests could act to pair neutral activities to become conditioned reinforcers. If embedded interest interventions can assist to not only expand skills, but also interests, individuals with ASD will be able to access new communities of reinforcement that can act to further expand their skills.

### Implications for Practitioners

There is a dynamic literature-base of interventions that embed interests into the learning environment for individuals with ASD. Given the variability in outcomes, the preliminary evidence suggests that practitioners should implement this intervention with caution and procedural modifications as necessary. Practitioners can compare an embedded interest condition to a no embedded interest condition using a withdrawal design or an alternating treatments design to determine effects within students. They can also look at studies included in this article to evaluate outcomes for participants with similar characteristics as their target student. If a target student exhibits high levels of RRBIs, stereotypical behaviors can be measured alongside skill targets to determine if this may be preventing success of the intervention strategy. Perhaps, moderate interests could be considered if embedding perseverative interests does not contribute to improved target outcomes.

When this intervention is applied alongside evidence-based practices, additive effects can occur. Engaging students with ASD is a challenging, but necessary aspect of designing effective educational programs. Embedding interests can be considered as a strength or interest-based approach to treatment. By incorporating what is important to individuals with ASD into their learning environments, they may become more inclined or motivated to engage in activities that are important to members of their community. As a result, individuals with ASD may experience more success with their social integration and learning outcomes.


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